

When do climate investments create jobs? The conditions behind employment creation

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1. Introduction

Low- and middle-income countries face a well-known bind: they need rapid economic growth and employment creation, and they need to decarbonize. Climate mitigation investment is often presented as a way to serve both goals – the claim that green spending creates more jobs per dollar than fossil fuel alternatives has become a staple of policy advocacy. But the evidence behind this claim rests on point estimates derived from fixed assumptions about what decarbonization costs, where the money goes, and how labor-intensive the resulting activities are. These assumptions vary enormously across countries and over time, and small changes in them can flip the employment arithmetic. A solar installation program that creates thousands of jobs when panels are manufactured domestically may create very few when they are imported. An energy efficiency retrofit that is labor-intensive at current wages may become capital-intensive as the sector matures. The existing literature tells policymakers that climate investment can create jobs; it does not tell them under what structural conditions that outcome holds, or when it breaks down. This paper addresses that question directly.

Using fixed cost structures from the literature for new green technologies can be misleading because those costs are often highly context specific rather than universally transferable. A technology that appears cost-effective in one setting may perform very differently elsewhere due to variation in input prices, labor costs, maintenance capacity, infrastructure compatibility, financing conditions, regulatory environments, and local resource availability.

Relying on a single fixed estimate can therefore obscure the real range of economic viability and lead the model to overstate precision. In doing so, it may miss how deployment outcomes change across places, over time, or under different future conditions, including learning effects, supply chain constraints, policy shifts, and operational uncertainties that are especially important in DMDU analyses.

The potential synergies between the climate agenda and economic growth have been at the center of discussions to whether a low carbon transition will contribute to net job creation, while reaching the required environmental goals.

Job creation is a key consideration when policymakers decide whether to pursue electricity production with 50% hydro and solar power or shift investments to expand wastewater treatment in urban areas by 40%. These decisions could be better informed if potential employment trajectories were assessed conditional on each country's economic structure, leveraging cost-structure variation to identify where co-benefits are most likely to materialize.

Why conditional findings matter (the "so what"):

This is key input for government and development agencies seeking to anticipate peaks in employment demand in specific sectors, align labor-market and training programs accordingly, and avoid sectoral bottlenecks.

Political economy hook:

The success of long-term climate strategies depends on their being understood and supported by both those responsible for implementing them and those affected by them (Waisman et al., 2019), yet decarbonization plans often lack guidance expressed in tangible pathways that stakeholders can comprehend and act upon (Kalra et al., 2024).

Contribution statement (also usable in Section 7):

The same transformation allocates investment differently across sectors depending on context and deployment assumptions, and that variation significantly affects employment outcomes. Scenario discovery identifies which allocation configurations produce substantial employment and which do not.

Helpful references to expand this section

- ILO (2018) “World Employment and Social Outlook” (#27) – the “24 million net new jobs” headline estimate. Open with this as the canonical claim, then show its fragility.
- Mealy et al. (2025) “Beyond Green Jobs” (#12) – the +8M to –80M range across studies. Use this spread to motivate the paper’s existence.
- Knudsen et al. (2025) “Five Frictions” (#15) – labor market barriers that prevent theoretical job creation from becoming realized employment.
- Saget et al. (2020) (#10) – the LAC-specific version of the canonical claim (+15M jobs). Ground the developing-country relevance early.

Possible discussion route

- Open with the policy tension, not the literature. The reader should understand why this matters for development before you tell them what other papers have done.
- The introduction should establish that this is a development economics question: structural transformation, labor absorption, import dependence – not just an energy policy question.
- State the research question cleanly after establishing the tension. Preview the framework briefly (SISEPUEDE → cost structures → I-O → scenario discovery) without getting technical.
- End with a roadmap paragraph. Keep it short.

2. Literature Review

2.1 Three modelling traditions in employment estimation

The employment effects of green investment, comparing the economic advantages of green spending against other investments, have received substantial attention. Systematic reviews summarize evidence comparing higher-carbon and low-carbon activities: the fossil fuel sector creates on average 5 per USD million invested compared to 15 jobs and 22 jobs per USD million for the renewable energy and energy efficiency technologies, respectively (Hanna et al., 2024; O’Callaghan et al., 2022). Studies have developed along three main methodological lines – Input-Output (IO), computable general equilibrium (CGE), and macro-econometric models – each with particular strengths and limitations. This paper explores a gap shared across these approaches: the treatment of cost structures of climate investments as known inputs when modeling job creation potential, despite the considerable uncertainty surrounding them. Whether by assuming fixed cost breakdowns, deriving cost trajectories from a fixed initial structure, or projecting historical trends forward, each tradition produces employment multipliers that may not fully capture the variability of key characteristics and structural conditions affecting job creation across contexts. The limitation becomes more consequential when economic sectors are expected to undergo fundamental restructuring in response to climate policies – adopting new practices and technologies rather than making incremental adjustments.

Input-output (IO) frameworks, including national and multi-region Input-Output (MRIO) models¹, have been increasingly used to estimate job creation potential in the low-carbon transition (Bücker et al., 2025; San Miguel et al., 2024; Černý et al., 2021; Montt et al., 2018). The framework traces how a dollar of exogenous final demand (e.g., new investment) propagates through inter-industry supply chains – translated into sectoral gross output and intermediate input requirements – capturing direct and indirect job creation at fine sectoral and regional resolution. Nonetheless, several studies in this tradition assume, at least partially, fixed cost breakdowns of green technologies or mitigation policies when constructing demand vectors (Bücker et al., 2025; Bakry et al., 2023; Ram et al., 2020; Garrett-Peltier, 2017)². This is a fragile assumption, as cost structures are subject to considerable uncertainty and may differ over time as technologies mature and across contexts. Employment multipliers can be a misleading measure of economic benefit when they do not account for variation in key factors such as the import shares of major components, the labor intensity of production, and prevailing wage levels (Agan and Balciar, 2022; Perrier and Quirion, 2018)³. This is also an issue in dynamic settings, where capital and operational expenditure trajectories remain dependent on the initial cost structure chosen.

¹ Grounded in national accounts and official statistics, MRIO databases (e.g., GLORIA, Eora) are increasingly used for empirical analysis as they provide a time-series of high resolution on interindustry and trade linkages across global supply chains, in addition to environmental and socio-economic extensions.

² Because current IO tables do not explicitly identify many green technologies or investment activities, such as renewable energy industries (e.g., solar, bioenergy), methodological extensions typically estimate employment multipliers by constructing weighted averages of existing industries represented in the I-O tables (Garret-Peltier, 2017; Pollin, 2016). Weights and industries involved usually rely on fixed cost structures from survey data and the climate-policy literature.

³ In more traditional setups, IO model results are better suited to analyzing short-term policy impacts because these models have limited ability to capture general equilibrium effects (e.g., adjusting wages, factor substitution), assume

Computable general equilibrium (CGE) models go further by allowing prices, wages, and factor substitution to adjust, imposing optimization and market clearing; some extensions produce net employment estimates that account for displacement and imperfect labor mobility (Mealy et al., 2025; Castellanos and Heutel, 2024; Fragkos & Fragkiadakis, 2022; Chateau et al., 2018; Bulavskaya and Reynès, 2018; Lofgren and Cicowiez, 2017). Despite these methodological developments, investments in green technology and mitigation actions are still modeled using cost structures treated as fixed parameters rather than as sources of uncertainty. CGE employment results are sensitive to production function calibration and closure assumptions (Castellanos and Heutel, 2024; Hafstead et al., 2022). Inaccurate capital-labor shares can systematically distort production function calibration and factor substitution patterns, thereby biasing simulated labor-demand responses (Bolarinwa, 2024). Similarly, when the input cost structures of green technologies and climate investments do not reflect country-specific import dependence and domestic supply capacity, models may misallocate demand across domestic industries and thereby overstate or understate employment gains. For instance, in the case of solar photovoltaic (PV) deployment, domestic employment effects depend in part on the extent to which panels and electrical equipment are imported rather than produced domestically, which can shift labor demand away from manufacturing and toward installation, project development, maintenance, and other local service activities.

A third strand, macro-econometric models such as energy-environment-economy (3E) frameworks (e.g., E3ME, NEMESIS), estimates dynamic adjustment from historical data in terms of key economic variables (GDP, employment, investment) over time, and has been applied particularly to economic analysis of European energy and climate policy (Alexandri et al., 2024; Ortega et al., 2020; Lehr and Lutz, 2019; Pollitt et al., 2015). These models are demand-led in the short to medium term, and combine national accounting identities with estimated behavioral equations to assess how demand shocks – policy shocks, new investment – affect sectoral output, which feeds into the labor demand through sectoral production functions and wage-setting equations, avoiding the market clearing assumptions of standard CGE models (Cambridge Econometrics, 2022; Akcigit et al., 2022). A potential weakness is that core behavioral relationships are estimated from historical time series, embedding in the model's coefficients sectoral interdependencies (e.g., labor-output elasticities) that may be less reliable for forward-looking policy analysis. When applied to sectors whose cost and input structures are rapidly changing, these coefficients are themselves a source of uncertainty in employment estimates.

2.2 Beyond cost structures: uncertainty and conditional findings

As discussed in the previous section, each tradition of estimating the job creation potential of green investments embeds fixed cost-structure assumptions and historical relationships in its models, but none treats them as uncertain. Although not directly linked to employment outcomes, uncertainty in cost trajectories has been documented in the low-carbon technology literature. Renewable energy technology costs in energy-economy models, such as integrated assessment models (IAMs), have been shown to be consistently overestimated relative to historical data, largely due to ad hoc assumptions – for instance, on adoption and diffusion rates in mitigation pathways (Achakulwisut et al., 2023; Way et al., 2022; Mercure et al., 2016). Beyond aggregate cost levels, a further source of uncertainty is that the composition of cost reductions can also shift over time, moving from low-level mechanisms (e.g., efficiency gain or reduced

perfectly elastic labor supply, and often rely on assumptions that become less plausible over the medium to long run (Mealy et al., 2025; Oosterhaven and Hewings, 2021).

material usage) in early stages to R&D and economies of scale in later ones (Kavlak et al., 2018; Rubin et al., 2015). Both the level and the internal composition of costs are affected as technologies and processes evolve, yet the climate employment literature treats them as known inputs.

Most employment estimates from climate investments are produced under a single cost configuration per action, with authors typically acknowledging cost-structure uncertainty in their limitations rather than addressing it analytically. A few studies went further by testing multiple configurations, but still without systematic exploration of the parameter space. Ram et al. (2022) estimate dynamic employment factors that decline as energy technologies mature, capturing changes in regional labor intensities over time. Nonetheless, the analysis is restricted to a single scenario with no alternative cost assumptions and no sensitivity tests on parameters (e.g., export-import dynamics). Černý et al. (2021) estimate employment effects from higher renewable penetration in the European Union electricity generation, varying deployment scenarios (100% renewables vs. baseline) but holding cost breakdowns fixed, with trajectories drawn from a single source (Carlsson et al., 2014).

Muller et al. (2024), modeling employment from green hydrogen infrastructure, find statistically significant differences in employment outcomes across 160 investment allocations that vary technology components and domestic supply shares. Yet the allocations draw on existing literature-based mappings, some outdated, and the analysis does not identify which conditions separate high from low employment outcomes. In the U.S. power sector, Bücker et al. (2025) assume a fixed spending breakdown for energy technologies, allowing unit costs to vary but not how each dollar is allocated across sectors. The authors explicitly acknowledge that this choice directly affects labor demand estimates, yet do not systematically map the conditions under which employment outcomes are favorable. Across these studies, cost-structure parameters are either held fixed or tested in discrete alternatives without systematic exploration of the parameter space. Where variation exists, no study identifies which combinations of conditions determine whether employment co-benefits materialize.

Recent work has called for country-specific, scenario-dependent analysis of labor demand in the green transition (Mealy et al., 2025). To our knowledge, no study has systematically explored which cost-structure configurations – at which implementation levels – separate cases where climate investments create substantial employment from those where they do not. This is a problem well suited to decision making under deep uncertainty (DMDU). The DMDU literature defines deep uncertainty as the condition where parties "*do not know or agree on the prior probability distributions for input parameters*" to their models (Bryant and Lempert, 2010). The variation in cost-structure assumptions documented above – across countries, technologies, and time – meets this condition. When no single parametrization can be validated, exploratory modeling offers an alternative: testing a wide range of plausible input combinations and tracing how outcomes respond (Lempert, 2025). Scenario discovery algorithms then identify which combinations of uncertain parameters separate favorable from unfavorable outcomes (Lempert, 2025; Bryant and Lempert, 2010). This approach has been widely applied to climate policy design (Kalra et al., 2024; Groves et al., 2020; Groves et al., 2019), including under technology cost uncertainty (Workman et al., 2024), but not to the employment outcomes of climate investments.

The cost-structure parameters that determine employment outcomes domestically – labor intensity, import shares, technology costs at varying deployment scales – cannot be characterized by stable probability distributions, particularly in settings where domestic industrial capacity and trade dependence are evolving.

Beyond single policies with fixed assumptions, no study examines how portfolios of climate actions at different implementation levels interact in their employment effects. A country doesn't implement one policy at full scale: it implements many at partial scale, and their joint employment effect may differ from the sum of individual estimates (due to shared supply chains, labor market competition between simultaneous deployments, or complementary domestic capacity building).

This is a problem well suited to the tradition of decision making under deep uncertainty (DMDU). The cost-structure parameters that determine employment outcomes – capital intensity, import shares, technology costs at varying deployment scales – cannot be characterized by stable probability distributions, particularly in developing countries where markets are thin, industrial capacity is uncertain, and structural change is rapid. Robust Decision Making (RDM), one class of DMDU methods (Marchau et al., 2019), systematically explores the consequences of assumptions across a wide range of plausible futures, then uses algorithmic methods to identify the combinations of conditions that distinguish favorable from unfavorable outcomes (Lempert, 2025). Rather than relying on predictions under specific assumptions, RDM produces a map of vulnerabilities: the structural conditions under which a policy succeeds or fails.

As discussed in section 2.1, each tradition of estimating the job creation potential of green investments embeds cost structure and historical coefficient assumptions in their models and results, but none treats them as uncertain. An additional gap identified is that studies haven't systematically explored how employment outcomes respond to varied implementation levels of climate policies individually and between policy packages. This is particularly relevant to inform national decarbonization plans and the formation of tailored policy packages, which should be explicit about the integration of socio-economic objectives – job creation, reduction of inequalities – to be policy-relevant (Waisman et al., 2026).

to improve guidance to policymakers who are required to balance current social needs of the population (e.g., job creation, public services) and the climate agenda, but lacking full depth or inputs to better assess current options available. On top of designing climate policies leading to net zero – pursuing electricity production with 50% hydro and solar power or shift investments to expand wastewater treatment in urban areas by 40% - those decisions could be better informed if potential job creation trajectories could be considered, leveraging each country's economic structures to maximize the co-benefits of climate policies. This goes in line with showing tangible economic benefits of climate policies, gathering the required political support required to sustained implementation, while meeting short and medium-term goals of government agencies, balancing climate and development goals (REFERENCE).

Job creation and destruction are one of the key considerations policymakers take into account when deciding whether to pursue electricity production with 50% hydro and solar power or shift investments to

expand wastewater treatment in urban areas by 40%, for example. This goes in line with showing tangible economic benefits of climate policies, gathering the required political support required to sustained implementation, while meeting short and medium-term goals of government agencies, balancing climate and development goals (REFERENCE).

Existing employment estimates do not explore how outcomes respond to variation in cost-structure assumptions. The employment generated by a given climate investment depends on whether equipment is produced domestically or imported, whether installation is labor-intensive or capital-intensive, and how technology costs evolve with deployment scale – factors that vary substantially across countries and over time. Yet the literature produces one estimate per intervention under a single assumed cost configuration, without examining how outcomes change as these structural conditions vary.

Even where sensitivity to individual parameters has been examined, the literature does not identify which combinations of structural conditions – and at what thresholds – determine whether a climate investment generates substantial or negligible employment. Employment outcomes depend on interactions among domestic production capacity, labor intensity, and technology costs; identifying the joint configurations that produce favorable results requires systematic exploration of the multi-dimensional parameter space, a step no existing study undertakes.

That imports, labor intensity, and wage levels influence employment multipliers is well established – cross-country differences in these structural characteristics already produce different estimates for the same investment when processed through multi-region input-output models. What remains unexplored is how employment estimates change when cost structures themselves shift – as they do when technologies mature, deployment scales up, and learning reduces equipment costs relative to installation. At low deployment levels, a climate investment allocates proportionally more to capital-intensive equipment; at higher levels, learning-driven cost reductions shift the cost structure toward installation and services. These dynamics mean that the employment generated by a given intervention depends not only on the receiving economy's structure but on the scale and stage of deployment – a dependence that static cost-structure assumptions cannot capture.

Even where the relationship between cost structures and employment is acknowledged, no study identifies the conditions – across deployment levels, cost configurations, and country contexts – under which climate investments generate substantial employment rather than negligible returns. A systematic exploration of how employment responds to simultaneous variation in deployment scale, cost-structure composition, and country-specific economic structure has not been undertaken. The literature cannot distinguish climate investments whose employment co-benefits hold across plausible configurations from those whose co-benefits depend on specific conditions that may not obtain in a given context.

That structural characteristics – import dependence, sectoral labor intensity, wage levels – influence employment multipliers is not in dispute; the models discussed in the previous section already embed those characteristics and produce country-specific estimates taking them into account. An additional gap identified is examining how employment estimates change when the cost structure of a climate investment shifts with deployment. As technologies scale up over time, reduced equipment costs relative to

installation are expected due to learning effects, altering the composition of expenditure across sectors (REFERENCE). In this sense, the same climate intervention at different deployment levels requires different investment levels and channels it differently to sectors of the economy. Studies have incorporated the impact of technology deployment levels in job creation in the energy sector (REFERENCE), yet this hasn't been done considering climate policies

through different parts of the economy – yet the literature estimates employment at a single assumed scale under a fixed cost breakdown, treating as static a structure that is, by its nature, evolving.

Even where the dependence of employment on cost-structure assumptions is acknowledged, no study identifies the conditions – across deployment levels, learning trajectories, and country contexts – under which climate investments generate substantial rather than negligible employment. The interaction between deployment scale, cost-structure evolution, and a country's sectoral composition determines employment outcomes in ways not derivable from inspecting individual factors in isolation. The literature can report that multipliers differ across contexts; it cannot specify the configurations under which employment co-benefits are robust, fragile, or unlikely to materialize.

No existing study systematically explores how employment outcomes respond to variation in cost-structure assumptions – whether across implementation levels of individual climate policies or across combinations of policies deployed simultaneously. In practice, countries do not implement a single mitigation action at full scale under known conditions; they implement multiple actions at partial scale, where costs depend on deployment trajectories, supply chain development, and interactions between concurrent investments. Yet the literature produces one employment estimate per action under one assumed configuration, leaving unexplored how outcomes vary across plausible structural conditions.

This is the domain of deep uncertainty: situations where parties to a decision cannot agree on the probability distributions governing key parameters or on the appropriate model linking actions to consequences (Lempert, 2025). Robust decision making (RDM), a class of decision making under deep uncertainty (DMDU) methods (Marchau et al., 2019), offers a framework for policy analysis under these conditions. Rather than reporting point estimates conditional on fixed assumptions, RDM systematically explores consequences across plausible futures and uses scenario discovery algorithms to identify the conditions under which a policy outcome is favorable or unfavorable (Lempert, 2025). The output is not a prediction but a map of conditions – those under which employment co-benefits materialize and those under which they do not.

Existing studies have connected climate policies to estimated job creation potential at the sector and occupational levels. For instance, Bücker et al. (2025) explores cyclical patterns of job creation in the U.S. power sector, tracking the transition into the different phases of climate investment - renewable infrastructure expansion, slow down once large projects conclude, and a steady-state job creation level. However, no study to the best of our knowledge has systematically explored how employment outcomes respond to varied implementation levels of a specific climate policy or a combination with different implementation levels. Development goals, such as job creation and destruction, are key considerations policymakers take into account when deciding whether to pursue electricity production with 50% hydro and solar power or shift investments or expand wastewater treatment in urban areas by 40%, for example.

This is key input to government and development agencies that can anticipate peak in employment demand in specific sectors, aligning labor-market and training programs, and avoiding sectoral bottlenecks.

The success of long-term climate strategies depends on their being understood and supported by both those responsible for implementing them and those affected by them (Waisman et al., 2019), yet decarbonization plans and climate research findings often lack guidance expressed in tangible and actionable pathways that stakeholders, such as policymakers and private sector, can comprehend and act (Kalra et al., 2024). An additional gap explored in this paper is that similarly, estimated job creation potential from climate policies are also not expressed in the context of tangible actions and lack traceability from specific mitigation actions to employment outcomes.

Existing studies lack traceability from specific mitigation actions to employment outcomes. They connect broad sectors or investment categories to jobs, but not the specific decisions that policymakers face – whether to pursue rooftop solar at 30% penetration or industrial efficiency upgrades at 50% penetration, and what each choice implies for labor demand.

The treatment of cost structures as known inputs, documented across all three modelling traditions in the preceding section, generates two downstream limitations. First, because cost parameters are fixed, no study systematically explores how employment outcomes respond to variation in these parameters - different implementation levels, deployment scales, import configurations, or learning trajectories. The same mitigation action modeled under alternative cost-structure assumptions can produce substantially different employment estimates, yet the literature reports a single estimate per action under one assumed configuration. Second, even if studies varied cost-structure parameters – running multiple scenarios or sensitivity analyses – this alone would not identify the conditions under which employment co-benefits materialize. A distribution of outcomes is not a conditional finding. Identifying which combinations of structural parameters (capital intensity thresholds, import share ceilings, minimum learning rates) separate substantial from negligible employment creation requires an additional analytical step: systematic exploration of the multi-dimensional parameter space paired with algorithms that extract decision-relevant rules from the resulting variation. No existing study performs this step. The consequence is that the literature cannot offer policymakers conditional guidance – it cannot say under which cost-structure configurations a given climate investment is likely to generate significant employment, nor how those configurations vary across country contexts.

What I vary vs. what I take as given:

- Varied: The demand vector – the sectoral allocation of each investment dollar. This reflects assumptions about domestic vs. imported content, capital vs. labor components, and technology costs at different deployment scales. The same climate action can route investment to different sectors depending on these upstream assumptions.

- Fixed: The I-O coefficients from GLORIA – jobs per unit of output, inter-industry linkages, sectoral import rates. These are taken as given for each country.

The implication:

The paper argues that cost-structure uncertainty matters for employment estimates broadly (Section 2.1 makes this case across I-O, CGE, and macro-econometric traditions). But the experimental design addresses only part of that uncertainty – the demand vector. The embedded I-O coefficients are also uncertain (the Uganda/Libya anomaly demonstrated this directly), but varying them is outside the scope of this design.

How to present this honestly:

- The contribution is: "the same transformation allocates investment differently across sectors depending on context and deployment assumptions, and that variation significantly affects employment outcomes." Scenario discovery then identifies which allocation configurations produce substantial employment.
- The limitation is: the I-O structure itself (labor productivity, inter-industry linkages, sectoral import rates) may not reflect a given country accurately, but is taken from GLORIA as given.
- This is a stated limitation, not a hidden one – and it's shared with every I-O study in the literature.

Similarly, estimated job creation potential from climate policies are also not expressed in the context of tangible actions and exploring different implementation levels appear disconnected from the priorities of policymakers actionable steps and may not fully capture national circumstances and characteristics, conditions that are key for the implementation by those typically responsible, such as government agencies and the private sector (Kalra et al., 2024; Gambhir et al., 2022). An additional gap explored in this paper is that similarly, estimated job creation potential from climate policies are also not expressed in the context of tangible actions and exploring different implementation levels

Instead, results are typically translated as the gross or net impact of demand shocks (e.g., new investment) into job numbers by sector.

What I want to say: employment estimates from climate policies connect broad sectors or investments categories lacking further guidance on traceability, meaning at what implementation level each policy should be pursued or the best combination of policies at varied implementation levels

or combinations of policies with varied implementation levels to reach specific and tangible development goals,

Existing studies lack traceability from specific mitigation actions to employment outcomes. They connect broad sectors or investment categories to jobs, but not the specific decisions that policymakers face – whether to pursue rooftop solar at 30% penetration or industrial efficiency upgrades at 50% penetration, and what each choice implies for labor demand.

Employment effects point estimates are produced not necessarily bringing to the discussion the conditions under which those effects can vary

Third, and as a consequence of the first two, the literature produces point estimates of employment effects rather than identifying the conditions under which those effects are large, small, or negative. Policymakers receive a number; they do not receive a map of what makes that number reliable or fragile.

The cost structure parameters that determine employment outcomes – labor intensity, import shares, learning rates⁴, technology unit costs – and how they could evolve in the future are deeply uncertain. Uncertainty of those parameters stem from multiple sources, such as technology shocks impacting labor share in production, price and trade shocks affecting imports, technology maturity and scale of deployment impacting technology costs. Those are a few examples of how the cost structure parameters can be affected by market developments and institutional conditions, ultimately affecting employment, that cannot be characterized by stable probability distributions, and should not be treated as known.

Mealy et al. (2025) “Beyond Green Jobs” (#12) – directly articulates all three gaps. The quote about estimates ranging from +8M to –80M is your best opening for Gap 3.

- Bucker et al. (2025) (#5) – their explicit acknowledgment that “cost breakdown of energy technologies... affects labor demand estimates directly” but they don’t systematically explore the space.
- Way et al. (2022) (#18) – strongest evidence that cost trajectories are deeply uncertain. Probabilistic forecasts outperform expert judgment. Anchors Gap 2.
- Cerny et al. (2022) (#4) – holds cost structures constant over 35 years. Acknowledges this is “an uncertain factor in the long run” but doesn’t address it.
- Knudsen et al. (2025) “Five Frictions” (#15) – five labor market barriers that create deep uncertainty about whether theoretical jobs become realized jobs.

Way et al., 2022:

- Future energy system costs will be determined by a combination of technologies that produce, store, and distribute energy
- The combination of exponentially decreasing costs and rapid exponentially increasing deployment is different from anything observed in any other energy technologies in the past and positions these key green technologies to challenge the dominance of fossil fuels within a decade.
- How likely is it that clean energy technology costs will continue to drop at similar rates in the future? Under what conditions will this happen, and what does this imply for the overall cost of the green energy transition? Is there a path forward that can get us to net-zero emissions cheaply and quickly?
- The second is Wright’s law, which predicts that costs drop as a power law of cumulative production. This relationship is also called an experience curve or learning curve

⁴ Learning rate is a parameter used to represent the percentage reduction in a technology’s unit cost that occurs when deployment/installed capacity doubles (Rubin et al., 2015).

Three limitations run through this literature, cutting across methodological traditions. First, existing studies lack traceability from specific mitigation actions to employment outcomes. They connect broad sectors or investment categories to jobs, but not the specific decisions that policymakers face – whether to pursue rooftop solar at 30% penetration or industrial efficiency upgrades at 50% penetration, and what each choice implies for labor demand. Second, the cost structures that determine how investment translates into sectoral spending are treated as fixed parameters rather than as sources of uncertainty. The share of investment that goes to capital equipment versus labor, or to imported versus domestically produced inputs, is typically assumed known and constant – even though these shares vary across countries, change as technologies mature, and depend on policy choices like local content requirements. Third, and as a consequence of the first two, the literature produces point estimates of employment effects rather than identifying the conditions under which those effects are large, small, or negative. Policymakers receive a number; they do not receive a map of what makes that number reliable or fragile.

The cost-structure parameters that determine employment outcomes – capital intensity, import shares, learning rates, technology unit costs – are not merely uncertain in the statistical sense. Their values depend on policy choices, market developments, and institutional conditions that cannot be characterized by stable probability distributions, particularly in developing countries where data is scarce, markets are thin, and structural change is rapid. This is the domain of deep uncertainty: situations where analysts cannot agree on the appropriate models, probability distributions, or even the relevant variables. The Robust Decision Making tradition, developed to support policy analysis under these conditions, offers a framework that inverts the standard modeling approach. Instead of selecting a best-estimate scenario and reporting results, it evaluates outcomes across a wide range of plausible futures, then uses algorithmic methods – principally the Patient Rule Induction Method (PRIM) – to identify the combinations of conditions that distinguish favorable from unfavorable outcomes. The output is not a prediction but a map of vulnerability: the structural conditions under which a policy succeeds or fails.

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Possible discussion route

- State each gap crisply and then provide evidence that it's real. Don't just assert – show that prominent papers acknowledge these limitations without addressing them.
- Gap 2 is where your development economics positioning is strongest. In developing countries, import dependence, domestic manufacturing capacity, and labor market structure are the central determinants of whether investment generates local employment. These are classic development economics variables.
- Gap 3 connects to the DMDU literature: the problem isn't insufficient modeling, it's that the question being asked is wrong. Instead of “how many jobs?” the question should be “under what conditions do jobs materialize?”
- End 2.2 by stating what would be needed to close these gaps. This sets up 2.3 and Section 3

The cost-structure parameters that determine employment outcomes – capital intensity, import shares, learning rates, technology unit costs – are not merely uncertain in the statistical sense. Their values depend on policy choices, market developments, and institutional conditions that cannot be characterized by stable probability distributions, particularly in developing countries where data is scarce, markets are thin, and structural change is rapid. This is the domain of deep uncertainty: situations where analysts cannot agree on the appropriate models, probability distributions, or even the relevant variables. The Robust Decision Making tradition, developed to support policy analysis under these conditions, offers a framework that inverts the standard modeling approach. Instead of selecting a best-estimate scenario and reporting results, it evaluates outcomes across a wide range of plausible futures, then uses algorithmic methods – principally the Patient Rule Induction Method (PRIM) – to identify the combinations of conditions that distinguish favorable from unfavorable outcomes. The output is not a prediction but a map of vulnerability: the structural conditions under which a policy succeeds or fails.

Helpful references to expand this section

- Bryant & Lempert (2010) (#14) – the operational description of PRIM and scenario discovery. Your primary methods citation.
- Lempert (2025) (#17) – updated DMDU methodological authority. The refined 4-step RDM framework.
- Marchau et al. (2019) (#13) – the definitive DMDU collection. Canonical framing.
- Kwakkel (2017) (#44) – Exploratory Modeling Workbench. Toolkit citation if you use it.

Possible discussion route

- Keep this section short and focused. You're not writing a DMDU review – you're explaining why scenario discovery is the right tool for the specific problem identified in 2.2.
- The key move: link deep uncertainty specifically to the developing-country context. In OECD economies, cost structures might be reasonably well-characterized. In Uganda or Libya, import shares, labor intensity, and domestic manufacturing capacity are genuinely unknown.

- Frame it as: “Given the nature of the uncertainty described in 2.2, what analytical approach is appropriate? One designed for exactly this kind of problem.”

3. The Analytical Framework

3.1 Overview and pipeline architecture

The analytical framework consists of six linked modules that together trace the employment consequences of climate mitigation investments from specific actions to aggregate and sectoral labor demand. The pipeline begins with the definition of mitigation transformations and their associated emissions pathways, proceeds through cost estimation with endogenous learning dynamics, translates investment requirements into sectoral demand vectors via input-output mapping, and culminates in an exploratory analysis that systematically varies cost-structure assumptions to identify the conditions under which employment outcomes are favorable or unfavorable. The distinguishing feature of this pipeline is end-to-end traceability: every employment estimate can be linked back to a specific mitigation action, at a specific penetration level, under a specific set of cost-structure assumptions, in a specific country context. Figure 1 presents the full workflow.

Traditionally, models take the decarbonization pathway as given and choose a combination of policies and technologies to be implemented mainly through cost minimization, particularly those approaching the power sector assessing electricity capacity and generation of different technologies (Bucker et al. 2025; Way et al., 2022). This disregard resource constraints, trade-offs, and synergies that are fundamental to help guide decision making in a context of deep uncertainty and challenging political economy issues (Waisman et al., 2026; Kane & Boulle, 2018). This is the domain of deep uncertainty: situations where parties to a decision cannot agree or do not know the prior probability distributions for key parameters or the best approach to model actions to consequences (Lempert, 2025). Decision making under deep uncertainty (DMDU) tradition offers a framework to support policy analysis under these conditions. Robust decision making (RDM) is one of a class of DMDU methods (Marchau et al., 2019) used to systematically explore the consequences of assumptions and expand the range of futures considered and scenario options to identify the main tradeoffs constraining or driving results (Lempert, 2025). Rather than relying on predictions under specific assumptions, the RDM outputs provide a map of vulnerabilities and conditions under which a policy succeeds or fails.

As deployment scales over time, the composition of expenditure changes: in early stages, climate investments are capital-intensive – dominated by equipment purchases and construction activities – while at higher penetration, operational expenditure rises in relative terms, channeling more spending toward maintenance and services, which are provided domestically. The same climate intervention at different implementation levels thus requires not only different investment levels but routes spending differently across sectors of the economy, with direct consequences for job creation. This is particularly relevant for national decarbonization plans, which integrate multiple interventions at varying scales and should be explicit about socio-economic objectives to be policy-relevant (Waisman et al., 2026).

Robust Decision Making (RDM), a class of DMDU methods (Marchau et al., 2019), explores how different plausible assumptions affect outcomes and identifies which combinations of conditions lead to high or low outcomes (Lempert, 2025). Rather than relying on predictions under a single set of assumptions, RDM

identifies the cost-structure configurations and implementation levels under which employment co-benefits hold or break down.

Helpful references to expand this section

- Your first paper (Frontiers in Climate, 2024) – the predecessor pipeline. Establish continuity and explain what this paper adds.
- Bucker et al. (2025) (#5) – their 4-step pipeline is the closest comparator. Note similarities and differences.

Possible discussion route

- The figure is critical here. Consider a flow diagram: Transformations → SISEPUEDE → Cost module → MINDSET → Employment → Experimental design → Scenario discovery.
- Emphasize what's new relative to Paper 1: the learning curve extension, the systematic uncertainty exploration, and the scenario discovery layer.
- Keep this section brief – it's an overview. The detail comes in 3.2–3.7.

3.2 Emissions pathway construction (SISEPUEDE)

The first module constructs emissions pathways for each country using SISEPUEDE, an integrated Python-based platform that models greenhouse gas emissions across sectors and transformation strategies. SISEPUEDE takes as input a set of mitigation transformations – specific technical or policy actions such as shifting electricity generation to solar, improving industrial energy efficiency, or electrifying urban transport – each defined at varying intensity levels from minimal to full implementation. For each transformation-intensity combination, the model produces a time series of emissions reductions and, critically, the associated investment requirements. These investment requirements are the entry point for the employment analysis: they define how much is spent, on what, and when.

Helpful references to expand this section

- SISEPUEDE technical documentation – cite the platform documentation or any published description.
- Your first paper (Frontiers in Climate, 2024) – for the original implementation and validation.

Possible discussion route

- Frame SISEPUEDE as a tool, not a contribution. The reader needs to understand what it produces, not its internal architecture.
- Briefly list the sectors covered (energy, transport, buildings, industry, AFOLU, waste) and the number of transformations.
- The key analytical point: different transformations at different intensities produce different cost profiles, which is what feeds the variation in employment outcomes.

3.3 Cost estimation and learning curve extension

The cost estimation module converts transformation-level activity into capital and operational expenditure streams. For each transformation, marginal capital requirements are estimated based on technology-specific unit costs, scaled by deployment level and deployment timeline. This paper extends the cost module used in the first paper by incorporating endogenous learning curves: technology unit costs decline as a function of cumulative deployment, following Wright's Law. The extension matters because the cost structure of a mitigation investment is not static – the same technology deployed at 10% penetration has a different cost profile than at 80% penetration, and the difference is not merely in scale but in composition. As unit costs fall with deployment, the relative shares of capital, labor, and imported components shift, altering the employment multiplier associated with each dollar invested. Learning rates are themselves uncertain, varying by technology and estimated with considerable noise, which makes them a natural dimension for exploratory analysis.

Learning curve → cost structure → sectoral allocation:

As technologies scale up, reduced equipment costs relative to installation – expected from learning effects – alter the composition of expenditure across sectors. The same climate intervention at different deployment levels requires different investment amounts and channels them differently through the economy.

Helpful references to expand this section

- Way et al. (2022) (#18) – empirical learning rates with uncertainty bounds. Solar PV: 19.8%, batteries: 25.3%, wind: 12.6%.
- Rubin et al. (2015) (#19) – comprehensive learning rate review for electricity technologies. Use for parameter ranges.
- Kavlak, McNerney & Trancik (2018) (#45) – decomposes PV cost reduction into contributing factors.
- Arrow (1962) (#66) – foundational citation for learning by doing.

Possible discussion route

- This is one of the paper's genuine contributions. But keep it focused on how learning curves feed into employment variation, not on the econometrics of learning curves.
- Development economics angle: learning curves imply early movers pay higher costs per unit (more labor-intensive, more job-creating early on), but also higher costs that strain fiscal space. Late movers benefit from lower unit costs but face higher import dependence if manufacturing has consolidated elsewhere.
- Be explicit about the functional form and parameterization.

3.4 Cost-structure translation to input-output analysis (MINDSET)

Investment requirements from the cost module must be translated into demand vectors compatible with the input-output framework – that is, into a specification of how each dollar of investment is distributed across sectors of the economy. This mapping is performed through MINDSET, which decomposes the cost of each transformation into spending on equipment, construction, professional services, and other intermediate inputs, then allocates these expenditures to the corresponding sectors of each country's I-O table. The mapping is the hinge of the entire pipeline: it is where assumptions about cost structure – the balance between capital and labor, between domestic production and imports, between specialized equipment and general construction – become determinative. Two countries investing the same amount in solar PV will generate different sectoral demand patterns depending on whether panels are imported or domestically manufactured, whether installation is performed by specialized firms or general contractors, and how the local supply chain for balance-of-system components is structured.

The translation from cost structures to employment estimates relies on MINDSET, a multi-regional input-output model developed by the World Bank to assess the sectoral employment effects of climate and development investments (MINDSET 1.0 Documentation, 2025). MINDSET operates on the GLORIA database, which covers 163 regions and 120 sectors with bilateral trade flows, supply-use transactions, and value added at basic prices (Lenzen et al., 2017; Lenzen et al., 2021). The model is demand-led: it assumes spare capacity and involuntary unemployment, such that investment shocks generate additional output without crowding out existing activity – an assumption better suited to developing-country contexts where underemployment is prevalent than general equilibrium frameworks that impose market clearing.

The central mechanism for this paper is the investment converter. When an investment enters MINDSET, a matrix of sector- and country-specific coefficients decomposes the expenditure into product-level demand across the 120 GLORIA sectors. The resulting demand is then disaggregated across producing countries using base-year trade shares from fixed capital formation data, splitting each product between domestic production and imports according to observed bilateral patterns. This produces an initial final demand vector that reflects both the composition of the investment and the country's existing trade structure. The Leontief inverse, then propagates this demand through inter-industry supply chains to yield total output requirements – direct and indirect – by country and sector. Employment follows from multiplying sectoral output changes by country-specific employment coefficients (jobs per unit of output), which vary substantially across contexts and account for differences in labor intensity and wage levels.

Two features of this setup matter for the experimental design. First, the investment converter coefficients determine which sectors receive demand – these are the cost-structure parameters that the paper treats as uncertain and varies across draws. Second, the trade-share disaggregation allocates demand between domestic and foreign producers at the sector level but cannot distinguish within-sector variation in domestic capacity for specific goods. A country may import 30% of its electrical machinery on average while importing 95% of the specific components required for solar PV deployment. The experimental design addresses this by adjusting effective domestic capacity as a separate parameter, beyond what the MRIO's base-year trade patterns capture.

Helpful references to expand this section

- Garrett-Peltier (2017) (#1) – her “synthetic industry” approach is the same basic idea: constructing spending profiles for clean energy as weighted averages of existing I-O sectors.
- Perrier & Quirion (2018) – the three channels (labor intensity, wages, imports) are exactly what the mapping determines.
- Muller et al. (2024) (#75) – shows ~half of employment effects leak abroad through supply chains.

Possible discussion route

- This is where the paper’s core argument becomes concrete. The mapping from investment to sectoral demand is where all the action is.
- Development economics framing: in developing countries, the sectoral composition of demand from climate investment depends on domestic industrial capacity, which is a structural feature of the economy.
- Be transparent about how the mapping was constructed for each country. Reviewers will scrutinize this.

3.5 Employment estimation

Given a demand vector for each transformation-country-cost-structure combination, employment is estimated through the standard Leontief demand-driven model. The total output required to satisfy the investment demand is computed as $x = (I - A)^{-1}d$, where A is the matrix of technical coefficients and d is the demand vector. Employment is then derived by applying sectoral labor coefficients – workers per unit of output – to the resulting output vector. The model captures both direct employment (in the sectors that receive the investment spending) and indirect employment (in upstream supplying sectors, through inter-industry linkages). It does not capture induced effects from household spending of earned wages, a deliberate scope choice that keeps the estimates conservative and avoids the most assumption-dependent component of employment multiplier analysis.

Helpful references to expand this section

- Your first paper (Frontiers in Climate, 2024) – same core method. Establish continuity.
- Perrier & Quirion (2018) – shows that I-O overestimates by 7–51% relative to CGE depending on the sector.
- Buchheim et al. (2020) – causal evidence that multipliers are state-dependent. Motivation for why I-O estimates should be treated as conditional.

Possible discussion route

- Keep this section concise. The method is standard Leontief.
- The critical choice to justify: why demand-driven I-O rather than CGE? The RDM framing transforms the I-O limitation into a feature by treating structural parameters as uncertainties to explore.
- Note the exclusion of induced effects and O&M employment. Scope choices, not oversights.

3.6 Experimental design

The scenario discovery analysis requires a structured exploration of the cost-structure parameter space. Five dimensions of uncertainty are varied: the capital-labor intensity of investment (the share of spending that goes to labor versus equipment), the domestic-import composition (the share sourced from domestic suppliers versus imported), technology unit costs (reflecting different assumptions about where on the cost curve each technology sits), learning rates (governing how quickly costs decline with deployment), and labor productivity in the receiving sectors. Parameter ranges for each dimension are drawn from the empirical literature and calibrated to reflect the variation observed across the seven countries in the sample. A Latin Hypercube Sampling strategy generates a large ensemble of parameter combinations, each of which is run through the full pipeline to produce a transformation-country-specific employment estimate. The result is a database of employment outcomes indexed by the cost-structure assumptions that produced them – the input to scenario discovery.

What is varied vs. fixed:

The experimental design varies the demand vector – the sectoral allocation of each investment dollar. This reflects assumptions about domestic versus imported content, capital versus labor components, and technology costs at different deployment scales. The same climate action can route investment to different sectors depending on these upstream assumptions. The I-O coefficients from GLORIA – jobs per unit of output, inter-industry linkages, sectoral import rates – are taken as given for each country.

Scope statement (also usable in 6.4):

The paper argues that cost-structure uncertainty matters for employment estimates broadly; Section 2.1 makes this case across I-O, CGE, and macro-econometric traditions. The experimental design addresses one component of that uncertainty: the demand vector. The embedded I-O coefficients are also uncertain but varying them is outside the scope of this design.

Helpful references to expand this section

- Way et al. (2022) (#18) – learning rate distributions with uncertainty bounds.
- Rubin et al. (2015) (#19) – learning rate ranges across electricity technologies.
- Perrier & Quirion (2018) – identifies which structural parameters most affect results.
- Bryant & Lempert (2010) (#14) – LHS sampling strategy for scenario discovery applications.

Possible discussion route

- A table of parameter ranges with literature sources is essential here.
- Development economics angle: parameter ranges should reflect developing-country realities. Import shares for solar in Uganda are structurally different from Germany.

- Explain the sampling strategy clearly. How many runs? How were ranges determined? Country-specific or common?
- This section is where reviewers will evaluate rigor. Be precise.

3.7 Scenario discovery

Scenario discovery takes the ensemble of employment outcomes generated by the experimental design and asks: what distinguishes the cases where employment effects are large from the cases where they are small? Using the Patient Rule Induction Method (PRIM), the analysis identifies regions of the cost-structure parameter space – “boxes” defined by threshold values on a subset of parameters – that concentrate cases with employment outcomes above or below policy-relevant thresholds. The algorithm iteratively peels away subsets of the parameter space to find combinations of conditions (e.g., import share below X and labor intensity above Y) that predict favorable employment outcomes with high coverage and density. The result is a set of interpretable conditional statements: climate investment in country Z generates substantial employment when conditions A, B, and C hold simultaneously. Classification trees complement the PRIM analysis by providing a different partitioning of the parameter space that may be more intuitive for policy audiences.

Helpful references to expand this section

- Bryant & Lempert (2010) (#14) – operational PRIM description. Coverage-density tradeoff.
- Lempert (2025) (#17) – updated RDM framework with MORDM extensions.
- Kwakkel (2017) (#44) – Exploratory Modeling Workbench if you use it as the computational platform.

Possible discussion route

- Define the classification criterion explicitly. What counts as “substantial” employment? A fixed threshold? Relative? This choice matters.
- Note that PRIM and classification trees may identify different conditions as important – this is informative, not a problem.
- Keep technical details for an appendix. Focus on what the method does and produces.

4. Country Cases: Seven World Bank CCDR Applications

4.1 The CCDR context and why this matters

The framework was applied to seven countries – Mexico, Belize, Uganda, Egypt, Morocco, Libya, and Bulgaria – as part of World Bank Country Climate and Development Reports (CCDRs). CCDRs are diagnostic assessments that examine how climate change and climate action intersect with a country’s development priorities, and they increasingly include analysis of the employment implications of proposed mitigation pathways. The application to real policy engagements is not incidental: it means the framework was stress-tested against the data realities, institutional contexts, and policy questions that matter in practice,

not in a controlled academic exercise. It also means the country set was determined by operational demand rather than analytical convenience – a feature that produces an eclectic but usefully heterogeneous sample spanning four continents, five income categories, and markedly different economic structures.

Helpful references to expand this section

- World Bank CCDD documentation – cite the relevant CCDDs for each country.
- Knudsen et al. (2025) “Five Frictions” (#15) – uses CCDD country cases to illustrate labor market barriers. Your paper does the same with a different set of countries.

Possible discussion route

- The “operational demand” justification is honest but insufficient for a journal paper. Show that the seven countries span the dimensions that matter: income level, manufacturing capacity, energy import dependence, labor market structure.
- Frame the eclecticism as a strength: if the framework works across Mexico and Uganda, that’s a stronger validation than seven similar OECD economies.
- Briefly explain what CCDDs are for readers outside the World Bank ecosystem.

4.2 Country profiles

Table X presents the seven countries along the structural dimensions that the analytical framework treats as determinative: income level, GDP composition, manufacturing value added as a share of GDP, energy import dependence, labor force participation, unemployment and underemployment rates, and the dominant emitting sectors targeted by each country’s CCDD. The variation is substantial. Mexico has a diversified industrial base with significant domestic manufacturing capacity; Belize is a small open economy with minimal manufacturing and high import dependence for capital goods. Uganda has among the lowest labor costs in the sample but also the least developed inter-industry supply chains, meaning that the indirect employment channel that drives much of the I-O result may be weaker there in practice. Bulgaria sits between the developing-country cases and the EU member states that dominate the existing evidence base. Egypt and Morocco have growing renewable energy sectors with emerging domestic manufacturing potential, while Libya’s economy remains dominated by hydrocarbons.

Helpful references to expand this section

- World Bank Development Indicators – for the structural comparison table.
- Saget et al. (2020) (#10) – LAC employment estimates (relevant to Mexico/Belize positioning).
- Ravillard et al. (2021) (#6) – LAC workforce data on non-qualified workers, skill gaps.
- Verpoort et al. (2024) (#73) – green value chain geography. How renewable cost heterogeneity reshapes where employment lands.

Possible discussion route

- A structured comparison table is essential. This is where you show the heterogeneity, not just assert it.

- The development economics framing: these structural characteristics are not just descriptive – they are the variables that your scenario discovery will reveal as determinative.
- Consider organizing countries into clusters based on structural similarity rather than geography.

4.3 Data sources and calibration

The analysis draws on national input-output tables, labor force statistics, technology cost data, and trade composition data for each of the seven countries. I-O tables are sourced from national statistical offices or multi-regional databases, with reference years varying by country. Sectoral labor coefficients are derived from labor force surveys and national accounts, adjusted for differences in sectoral classification across countries. Technology cost data are drawn from IRENA, IEA, and the SISEPUDE cost module, with learning curve parameters calibrated to Way et al. (2022) and Rubin et al. (2015). Data quality and availability vary significantly across the seven countries – a reality that the exploratory framework accommodates rather than ignores, since parameter uncertainty is the object of analysis rather than a nuisance to be assumed away.

Helpful references to expand this section

- Way et al. (2022) (#18) – learning rate calibration source.
- Rubin et al. (2015) (#19) – learning rate parameter ranges for electricity technologies.

Possible discussion route

- Be ruthlessly transparent about data limitations. A table listing data sources by country and variable would strengthen this section.
- The framing: data limitations in developing countries are part of the motivation, not a weakness. Exploratory analysis is designed for imperfect data contexts.
- Note assumptions about sectoral concordance across different I-O table vintages and classifications.

5. Results

5.1 Transformation-level employment estimates

The pipeline produces employment estimates for each mitigation transformation, at each intensity level, in each country, under each cost-structure parameterization. Before turning to the uncertainty exploration and scenario discovery – the paper’s primary contributions – it is useful to examine the baseline estimates to establish what the framework produces when cost structures are held at their reference values. Table Y presents employment multipliers (jobs per million USD invested) for selected transformations across the seven countries. Three patterns are immediately visible: first, the same transformation generates markedly different employment across countries, driven by differences in domestic supply chain depth and labor intensity; second, building-sector transformations (retrofits, efficient construction) tend to produce higher employment than energy-sector transformations (utility-scale solar, wind), reflecting the higher labor intensity of construction; third, the cross-country variation is large enough to suggest that country-level

structural conditions, not the choice of transformation, are the primary determinant of employment outcomes.

Helpful references to expand this section

- Your first paper (Frontiers in Climate, 2024) – compare findings to the LAC results.
- Garrett-Peltier (2017) (#1) – compare your multipliers to her US benchmarks.
- Cerny et al. (2022) (#4) – their EU27+UK results for cross-country variation comparison.

Possible discussion route

- Lead with the cross-country comparison, not transformation-level detail. The development economics contribution is about why the same investment has different employment consequences in different structural contexts.
- A heat map (transformations $\times$ countries, colored by employment multiplier) could be the central visualization.
- These are the starting point, not the punchline. The punchline is the scenario discovery in 5.3.

5.2 The role of cost-structure assumptions

Holding the country and transformation fixed, how much do employment estimates change as cost-structure assumptions vary? The answer, across the ensemble of experimental runs, is: substantially. Figure Z plots the distribution of employment outcomes for a selected transformation in a selected country across the full parameter space, showing a range from [X to Y] jobs per million USD. This variation is driven primarily by [the parameter(s) revealed as most influential]. The range of employment estimates within a single country and transformation, generated by varying cost-structure assumptions alone, is in many cases as large as the range across countries under fixed assumptions. This means that getting the cost structure right – or at least understanding how sensitive results are to cost-structure assumptions – is at least as important as getting the country-level I-O table right.

Helpful references to expand this section

- Way et al. (2022) (#18) – if learning rate variation is a major driver, cite their evidence that cost trajectories are deeply uncertain.
- Perrier & Quirion (2018) – if labor intensity and import shares dominate, cite their decomposition.
- Bucker et al. (2025) (#5) – their sensitivity analysis found T&D costs and imports as largest sources. Compare.

Possible discussion route

- This section bridges baseline results (5.1) and scenario discovery (5.3). It shows cost-structure uncertainty matters before showing what the algorithm finds.
- Use density plots or violin plots to show distributions. Avoid reporting only means and standard deviations.

- Development economics point: the parameters that drive the most variation (import share, domestic manufacturing) are precisely the variables that development policy can influence.

5.3 Scenario discovery: conditions for positive job creation

The scenario discovery analysis identifies the cost-structure conditions that separate high-employment from low-employment outcomes across the seven countries. The PRIM algorithm, applied to the ensemble of experimental runs with employment outcomes classified as above or below a policy-relevant threshold, identifies a set of interpretable boxes in the parameter space that concentrate favorable outcomes. [State the central finding when results are available.] Across countries, [describe cross-country patterns]. Within individual countries, the binding conditions differ: for [country X], the critical parameter is [parameter], while for [country Y], the outcome depends primarily on [different parameter]. The scenario discovery also reveals which parameters are not binding – dimensions of cost-structure uncertainty that, within the ranges explored, do not significantly affect whether employment outcomes are favorable.

Helpful references to expand this section

- Bryant & Lempert (2010) (#14) – for interpreting PRIM output (coverage, density, interpretability tradeoffs).
- Mealy et al. (2025) (#12) – they called for conditional analysis; you deliver it.
- Knudsen et al. (2025) (#15) – do the frictions they identify correspond to the conditions your scenario discovery surfaces?

Possible discussion route

- Structure as: (1) cross-country universal conditions, (2) country-specific conditions, (3) transformation-specific conditions.
- Present PRIM threshold rules both formally and in plain language.
- The development economics contribution crystallizes here: the conditions for employment creation are structural features of developing economies (import dependence, labor intensity, manufacturing capacity).

5.4 Policy-relevant typology

The scenario discovery findings can be synthesized into a typology of employment vulnerability: a classification of country-transformation combinations based on how robust their employment outcomes are to cost-structure uncertainty. At one end are robust cases – combinations where employment outcomes are positive across most of the parameter space, meaning that job creation potential is resilient to variation in cost-structure assumptions. At the other end are fragile cases – combinations where employment outcomes are highly sensitive to one or two parameters, meaning that a policy choice (such as local content requirements or domestic manufacturing subsidies) could determine whether the investment creates substantial employment or very little. Between these extremes are cases where employment outcomes are conditionally favorable, depending on specific structural conditions that differ by country. This typology is the paper's primary deliverable for policymakers: not a single number, but a map of when and why climate investments create jobs.

Helpful references to expand this section

- Mealy et al. (2025) (#12) – their “jobs-in-demand” and “jobs-at-risk” framework. Your typology operationalizes their conceptual proposal.
- Ganter et al. (2024) (#31) – even when aggregate effects are positive, distributional patterns depend on cost structures and geography.

Possible discussion route

- A 2×2 or matrix visualization (robustness × employment magnitude) could be the paper’s most memorable figure.
- Development economics framing: does the typology map onto recognizable development categories? Are low-income countries systematically “fragile”? Does manufacturing capacity predict robustness?
- This is where the paper shifts from analytical contribution to policy contribution. Make the typology actionable.

6. Discussion

6.1 What cost-structure levers can governments actually influence?

The scenario discovery identifies cost-structure parameters that determine employment outcomes. But for these findings to be useful, policymakers need to know which of these parameters they can actually move. Some are structural features of the economy that change only slowly – the depth of inter-industry supply chains, the skill composition of the labor force. Others are more directly responsive to policy: local content requirements can raise the domestic share of investment spending; procurement standards can favor labor-intensive construction methods; industrial policy can support domestic manufacturing of key components; trade policy can shift import shares. This section maps the scenario discovery findings onto the policy instruments available to each country, distinguishing between conditions that represent binding structural constraints and conditions that represent actionable policy levers.

Helpful references to expand this section

- Verpoort et al. (2024) (#73) – green value chain geography and the “renewables pull.” How policy choices about where to locate manufacturing reshape employment geography.
- Muller et al. (2024) (#75) – Saudi Arabia hydrogen. Import shares determine whether employment stays domestic or leaks abroad.
- Atik Gozkun & Orhangazi (2025) (#24) – Turkey green transition. Trade dependencies and employment multipliers.

Possible discussion route

- Frame around what governments in developing countries can actually do, given fiscal and institutional constraints.

- Distinguish short-run levers (procurement, local content) from long-run levers (industrial policy, workforce development).
- Development economics framing: industrial policy for green manufacturing is a live policy question in many developing countries. Your findings provide empirical guidance on when it matters for employment.

6.2 Implications for CCDR design and climate strategy

Current practice in country climate assessments – including the CCDRs that this framework was developed to support – typically reports employment co-benefits as single-scenario estimates. The analysis presented here suggests this practice understates the decision-relevant information available. A point estimate of “200,000 jobs from solar investment” tells a finance minister less than the statement “solar investment creates 150,000–250,000 jobs depending on import share, and the outcome depends critically on whether the country develops domestic panel manufacturing.” The shift from point estimates to conditional findings changes both what analysts report and what policymakers can act on. For future CCDRs and similar assessments, this implies integrating exploratory analysis into the standard workflow – not as a robustness check appended to the main results, but as the primary analytical lens.

Helpful references to expand this section

- Mealy et al. (2025) (#12) – they advocate moving beyond static green job counts toward scenario-conditioned analysis.
- Knudsen et al. (2025) (#15) – their call for incorporating frictions into macroeconomic modeling.

Possible discussion route

- Be concrete about what would change in CCDR practice. The institutional audience is World Bank teams and country counterparts.
- Development economics angle: CCDRs are development documents. If employment co-benefits are conditional on structural features, CCDRs should be explicit about those conditions.

6.3 Relationship to prior literature

The baseline employment estimates produced by this framework are broadly consistent with the existing literature: clean energy investments generate more jobs per dollar than fossil fuel investments, with the difference driven by higher labor intensity and stronger domestic supply chain linkages. Where this paper departs from prior work is in showing how large the uncertainty around these estimates is, and in identifying what drives it. Garrett-Peltier’s (2017) benchmark US multipliers fall within the range produced by this framework’s ensemble, but they represent a single point in a distribution whose width is determined by cost-structure assumptions she treated as fixed. Similarly, the LAC estimates of Saget et al. (2020) and the power-sector estimates of Bucker et al. (2025) are consistent with subsets of this paper’s ensemble – specifically, the subsets where cost-structure parameters are set near their reference values. The scenario discovery adds the dimension these studies lack: the conditions under which their estimates hold, and the conditions under which they break.

Helpful references to expand this section

- Garrett-Peltier (2017) (#1) – direct comparison of multiplier ranges.
- Bucker et al. (2025) (#5) – their sensitivity findings vs. yours.
- Saget et al. (2020) (#10) – +15M LAC jobs as a point within your distribution.
- Cerny et al. (2022) (#4) – EU27+UK results with zero uncertainty analysis.
- Callonnec et al. (2025) (#74) – CGE France results. Compare to your I-O results for Bulgaria.

Possible discussion route

- Structure as: (1) where your findings confirm existing estimates, (2) where they challenge or qualify them, (3) what your framework adds.
- Be fair to the literature. Previous studies were not wrong – they reported one slice of a larger space.

6.4 Limitations

Three categories of limitations constrain the interpretation of these results. First, the input-output framework is static and demand-driven: it does not model price adjustments, factor substitution, or general equilibrium effects. In labor markets with full employment, the I-O estimates represent an upper bound on job creation because they ignore crowding-out. In labor-surplus economies – several of the countries in this sample – the demand-driven assumption may be more appropriate, but the framework still does not model wage adjustments or the dynamics of labor reallocation. Second, the analysis estimates labor demand, not labor market outcomes. It does not address job quality, occupational composition, wage levels, gender composition, or the geographic distribution of employment – dimensions that matter substantially for whether employment gains translate into improved livelihoods. Third, data quality varies across the seven countries, with some I-O tables more disaggregated and recent than others. The exploratory framework treats parameter uncertainty, but it does not capture all forms of data uncertainty, particularly errors in the I-O tables themselves.

The I-O structure itself – labor productivity, inter-industry linkages, sectoral import rates – may not reflect a given country accurately, but is taken from GLORIA as given. This is a stated limitation, not a hidden one, and it is shared with every I-O study in the literature.

Helpful references to expand this section

- Perrier & Quirion (2018) – quantifies I-O overestimation relative to CGE (7–51%).
- Buchheim et al. (2020) – causal evidence that multipliers are zero in tight labor markets.
- Vona et al. (2018) (#50) or Curtis & Marinescu (2023) (#52) – job quality dimensions you’re excluding.
- Bowen et al. (2018) (#51) – definitional issues with “green employment.”

Possible discussion route

- The RDM reframing: static I-O assumptions are explored, not ignored. The experimental design varies the parameters that embody these assumptions.
- Be honest about what the paper cannot do. Reviewers will respect candor.
- Note that job quality, distributional effects, and transition costs are exclusions, not oversights. Point to future work.

7. Conclusions

This paper asks a question that the existing literature on climate mitigation and employment has not addressed directly: under what structural conditions do climate investments create jobs, and how do those conditions vary across countries at different stages of development? The answer, derived from applying an end-to-end analytical framework to seven World Bank CCDR country engagements and exploring cost-structure uncertainty through scenario discovery, is that employment co-benefits are real but conditional. The conditions are identifiable, country-specific, and in many cases responsive to policy. [State the 2–3 most important specific findings.] These findings suggest that the prevailing approach to assessing employment co-benefits – reporting point estimates under fixed cost-structure assumptions – systematically obscures the information that policymakers most need.

The same transformation allocates investment differently across sectors depending on context and deployment assumptions, and that variation significantly affects employment outcomes. Scenario discovery identifies which allocation configurations produce substantial employment and which do not.

Helpful references to expand this section

- No new references needed. The conclusion should draw on everything cited earlier.

Possible discussion route

- Restate the three contributions concisely: (1) an end-to-end pipeline with transformation-level traceability, (2) scenario discovery findings about cost-structure conditions for employment creation, (3) demonstrated applicability across seven structurally diverse countries.
- End with the development economics message: employment creation from climate investment is not automatic – it depends on the same structural features that define a country’s development status.
- Keep it short. One page maximum.

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